



SRI LANKA TECHNOLOGY CAMPUS
ASSIGNMENT 02
MINI PROJECT

Due Date: 23rd Nov. 2025

Module: Internet Technologies (CIT201)

Topic: Implementing a SOHO Network with Services

1. Project Goal

Design and configure a Small Office/Home Office (SOHO) network to provide local client PCs with automatically assigned IP addresses, internet access via NAT, and access to a local server using a friendly Domain Name (DNS).

2. Network Topology and Devices

Device Type	Quantity	Role	Course Concept Covered
Router (e.g., Cisco 4321)	1	Connects internal LAN to simulated WAN (Internet).	Routing, NAT
Switch (e.g., Cisco 2960)	1	Connects local client devices.	Hardware
Server (Generic)	1	Provides central network services.	Application Software/Services
End Devices (PCs/Laptops)	3-4	Client workstations to test services.	Hardware
Cloud/ISP (Optional)	1	Simulates connection to the public internet for NAT testing.	Internet Architecture

3. Step-by-Step Configuration Instructions

1. Step 1: IP Addressing and DHCP

- Configure the Server to run a DHCP service to automatically assign IP addresses, subnet masks, and a default gateway to all PCs/Laptops.
- Assign a static IP address to the Server.

2. Step 2: DNS and Web Services

- Configure the Server to run the DNS service.
- Create a DNS record to resolve a custom name (e.g., www.myoffice.com) to the Server's static IP address.
- Enable the HTTP service (Web Server) and modify the index.html file with a welcome message.

3. Step 3: Routing and NAT

- Configure NAT on the Router to allow internal (private) IPs to share a single public IP when accessing the external network.

4. Step 4: Security (Bonus Task)

- Configure Telnet or SSH on the Router for remote management from a PC.

4. Verification and Testing

1. Ensure that all PCs obtain IP configuration automatically (IP, Subnet Mask, Gateway, DNS).
2. Test connectivity using ping between PCs, the Router, and the Server.
3. Open a browser on a client PC and access the web page using the DNS name (www.myoffice.com) to verify functionality.

5. Additional Resources

For a detailed walkthrough, refer to the following videos:

Networking Mini Project-1 using CISCO Packet Tracer:

<https://youtu.be/mRyVd0lM5E4?si=rZtctrCGnDKr5ofp>

1. DHCP and Server Configuration
 - a. DHCP Server Setup - <https://youtu.be/ZCz9rRhzy4?si=IVyt-gP0TBPPD5cp>
2. Static and Dynamic Routing (EIGRP/OSPF)
 - a. Static Routing - https://youtu.be/vrf2016imtQ?si=vx-4_firEATcG52b
 - b. Dynamic Routing (EIGRP) - <https://youtu.be/jWVxt1esrBw?si=9jfs1BXPAt3d7GaN>
3. Network Address Translation (NAT)
 - a. NAT Configuration - https://youtu.be/HD1WaIN-fvE?si=i-H_ph84jaed12CO
4. Advanced Security (VPN)
 - a. VPN Setup - <https://youtu.be/mMhCtLlRfDM?si=vPLJu4BoTapFNX3->